In [1]:
```
import numpy as np
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier,plot_tree
import matplotlib.pyplot as plt
```

In [17]:
```
df = pd.read_csv("/home/backup/ml_datasets/OnlineRetail1.csv",encoding='cp1252')

df = df.dropna(subset = ['CustomerID'])
df = df[(df['Quantity']>0) & (df['UnitPrice']>0)]
df['TotalSpend'] = df['Quantity'] * df['UnitPrice']
customer_df = df.groupby('CustomerID').agg({'Quantity':'sum', 'UnitPrice':'mean', 'T

print(df.head(5))
```

```
  InvoiceNo StockCode                          Description  Quantity  \
0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER         6
1    536365     71053                  WHITE METAL LANTERN         6
2    536365    84406B       CREAM CUPID HEARTS COAT HANGER         8
3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE         6
4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.         6

        InvoiceDate  UnitPrice  CustomerID         Country  TotalSpend
0  01-12-2010 08:26       2.55     17850.0  United Kingdom       15.30
1  01-12-2010 08:26       3.39     17850.0  United Kingdom       20.34
2  01-12-2010 08:26       2.75     17850.0  United Kingdom       22.00
3  01-12-2010 08:26       3.39     17850.0  United Kingdom       20.34
4  01-12-2010 08:26       3.39     17850.0  United Kingdom       20.34
```

In [3]:
```
threshold=customer_df['TotalSpend'].median()
customer_df['HighValue']=(customer_df['TotalSpend']>threshold).astype(int)

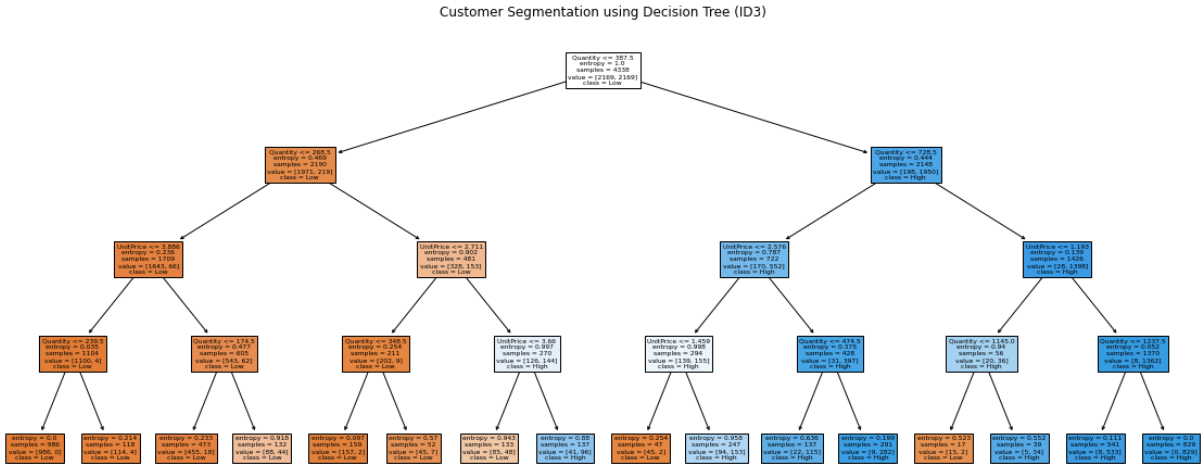
#Encode categorcical data
le=LabelEncoder()
customer_df['Country']=le.fit_transform(customer_df['Country'])
```

In [4]:
```
X = customer_df[['Quantity', 'UnitPrice', 'Country']]
y = customer_df['HighValue']
```

In [5]:
```
model = DecisionTreeClassifier(criterion='entropy', max_depth=4)
model.fit(X,y)
```

Out[5]:
```
DecisionTreeClassifier(criterion='entropy', max_depth=4)
```

In [16]:
```
plt.figure(figsize = (20,8))
plot_tree(model,
        feature_names = X.columns,
        class_names = ['Low','High'],
        filled = True)
plt.title("Customer Segmentation using Decision Tree (ID3)")
plt.show()
```



In [18]:

```
importance = pd.Series(model.feature_importances_, index = X.columns)
print("\n Feature Importance : \n", importance.sort_values(ascending = False))
```

```
 Feature Importance :
 Quantity     0.86527
UnitPrice    0.13473
Country      0.00000
dtype: float64
```

In [ ]: